Supplementary Information

DPGOK: A deep learning-based method for protein function prediction by fusing GO knowledge with protein features

Table S1. The number of edge for each GO relational subgraph on three sub-ontologigs.

Relations	MFO		BPO		CCO	
	Before propagating	After propagating	Before propagating	After propagating	Before propagating	After propagating
is_a	7815	46549	32600	242673	3044	15177
part_of	10	6101	10892	50346	3539	11356
positively_regulates	19	19	6226	6242	0	0
negatively_regulates	0	0	5601	5634	0	0

Table S2. Mean AUPR and P-values for GO Terms by Depth Group

Ontology	Methods	Depth < 4		Depth 4-6		Depth > 6	
		M_AUPR	P-value	M_AUPR	P-value	M_AUPR	P-value
MFO	BlastKNN	0.470	1.0e-4	0.560	1.4e-8	0.681	3.0e-4
	TALE	0.324	1.9e-10	0.354	2.0e-43	0.519	5.0e-12
	DeepGOZero	<u>0.535</u>	0.45	<u>0.579</u>	3.2e-2	<u>0.746</u>	0.92
	ATGO	0.446	2.0e-7	0.477	1.2e-26	0.578	2.5e-7
	DeepGO-SE	0.517	0.14	0.570	4.1e-6	0.719	0.02
	DPGOK	**0.545**	-	**0.615**	-	**0.758**	-
	TALE+	0.457	5.3e-7	0.516	2.8e-171	0.665	3.8e-5
	ATGO+	<u>0.505</u>	0.04	<u>0.586</u>	2.9e-4	<u>0.695</u>	1.8e-3
	DPGOK +	**0.559**	-	**0.640**	-	**0.778**	-
BPO	BlastKNN	0.245	1.4e-13	0.249	2.0e-43	0.295	2.5e-8
	TALE	0.098	3.7e-25	0.051	2.4e-235	0.067	1.0e-5
	DeepGOZero	0.231	1.0e-13	0.211	3.0e-71	0.249	2.1e-19
	ATGO	0.220	3.3e-17	0.177	4.9e-146	0.225	9.2e-32
	DeepGO-SE	<u>0.272</u>	1.6e-4	<u>0.260</u>	1.3e-11	<u>0.302</u>	5.3e-9
	DPGOK	**0.318**	-	**0.286**	-	**0.343**	-
	TALE+	0.273	1.0e-9	0.260	1.0e-28	0.306	9.1e-5
	ATGO+	<u>0.295</u>	3.0e-4	<u>0.267</u>	4.3e-21	<u>0.318</u>	2.9e-4
	DPGOK +	**0.333**	-	**0.308**	-	**0.371**	-
CCO	BlastKNN	0.344	4.3e-8	0.282	1.9e-16	0.510	2.5e-4
	TALE	0.126	3.8e-15	0.035	1.0e-43	0.182	4.7e-7
	DeepGOZero	0.320	8.64e-10	0.242	1.5e-27	0.497	1.0e-4
	ATGO	0.308	1.3e-8	0.219	1.9e-29	0.403	9.5e-7
	DeepGO-SE	<u>0.373</u>	2.0e-3	<u>0.395</u>	3.3e-7	<u>0.641</u>	0.02
	DPGOK	**0.431**	-	**0.429**	-	**0.720**	-
	TALE+	0.359	6.4e-5	0.308	7.6e-16	<u>0.531</u>	1.17e-3
	ATGO+	<u>0.383</u>	3.2e-5	<u>0.335</u>	1.7e-2	0.528	5.3e-4
	DPGOK +	**0.441**	-	**0.445**	-	**0.711**	-

Note: The best-performing value in each category is shown in bold, and the second-best is underlined. P-values are calculated relative to DPGOK.

Table S3. The median of AUPR for GO terms within each Depth group

Methods	Depth < 4			Depth 4-6			Depth > 6		
	MFO	BPO	CCO	MFO	BPO	CCO	MFO	BPO	CCO
BlastKNN	0.497	0.187	0.324	<u>0.583</u>	0.150	0.282	<u>0.809</u>	0.208	0.465
TALE	0.246	0.047	0.048	0.314	0.015	0.035	0.507	0.022	0.138
DeepGOZero	<u>0.506</u>	0.180	0.290	0.559	0.103	0.242	0.833	0.151	0.415
ATGO	0.345	0.149	0.262	0.502	0.069	0.219	0.567	0.095	0.494
DeepGO-SE	0.496	<u>0.224</u>	<u>0.369</u>	<u>0.583</u>	<u>0.170</u>	<u>0.395</u>	0.778	<u>0.207</u>	<u>0.704</u>
DPGOK	**0.583**	**0.288**	**0.400**	**0.668**	**0.202**	**0.429**	**0.859**	**0.294**	**0.824**
TALE+	0.444	0.222	0.323	0.506	<u>0.171</u>	0.308	0.726	0.244	0.518
ATGO+	<u>0.489</u>	<u>0.251</u>	<u>0.363</u>	<u>0.598</u>	0.169	<u>0.335</u>	<u>0.833</u>	<u>0.251</u>	<u>0.485</u>
DPGOK +	**0.583**	**0.294**	**0.399**	**0.700**	**0.236**	**0.445**	**0.892**	**0.314**	**0.799**

Note: The best-performing value in each category is shown in bold, and the second-best is underlined.

Table S4. Mean AUPR and P-values for GO terms within each IC group

Ontology	Methods	IC < 3		IC 3-5		IC > 5	
		M_AUPR	P-value	M_AUPR	P-value	M_AUPR	P-value
MFO	BlastKNN	0.573	9.5e-9	<u>0.622</u>	9.2e-4	0.526	6.6e-6
	TALE	0.393	4.0e-45	0.384	3.7e-28	0.394	9.1e-11
	DeepGOZero	0.582	0.01	0.598	0.14	<u>0.543</u>	8.7e-3
	ATGO	0.482	8.7e-28	0.435	6.5e-22	0.356	7.4e-16
	DeepGO-SE	<u>0.603</u>	3.6e-3	0.580	7.3e-6	0.527	5.8e-4
	DPGOK	**0.624**	-	**0.636**	-	**0.595**	-
	TALE+	0.534	6.1e-18	0.560	2.9e-11	<u>0.539</u>	2.0e-3
	ATGO+	<u>0.581</u>	5.8e-7	<u>0.629</u>	0.15	0.523	1.3e-6
	DPGOK +	**0.650**	-	**0.667**	-	**0.603**	-
BPO	BlastKNN	<u>0.257</u>	4.6e-57	<u>0.242</u>	1.8e-28	0.103	1.1e-18
	TALE	0.014	2.5e-297	0.014	1.7e-137	0.016	3.2e-37
	DeepGOZero	0.145	4.1e-78	0.113	1.1e-35	0.099	1.0e-11
	ATGO	0.108	6.0e-196	0.068	8.1e-102	0.030	3.7e-30
	DeepGO-SE	0.184	9.6e-15	0.167	1.5e-4	<u>0.278</u>	0.68
	DPGOK	**0.283**	-	**0.271**	-	**0.281**	-
	TALE+	0.266	7.2e-31	0.246	5.9e-18	0.122	3.2e-37
	ATGO+	<u>0.269</u>	2.9e-33	<u>0.248</u>	5.6e-25	<u>0.245</u>	2.6e-14
	DPGOK +	**0.303**	-	**0.290**	-	**0.301**	-
CCO	BlastKNN	0.362	8.9e-17	0.396	2.7e-9	0.379	8.0e-9
	TALE	0.046	3.9e-46	0.039	2.7e-28	0.053	5.3e-13
	DeepGOZero	0.316	2.5e-26	0.348	5.1e-12	0.403	7.9e-5
	ATGO	0.282	1.1e-26	0.263	5.2e-22	0.302	2.3e-9
	DeepGO-SE	<u>0.402</u>	3.7e-6	<u>0.427</u>	2.6e-5	<u>0.477</u>	0.52
	DPGOK	**0.468**	-	**0.482**	-	**0.503**	-
	TALE+	0.355	9.0e-16	<u>0.410</u>	1.3e-6	0.404	1.4e-4
	ATGO+	<u>0.391</u>	6.6e-12	0.397	1.2e-8	<u>0.429</u>	3.4e-7
	DPGOK +	**0.476**	-	**0.489**	-	**0.510**	-

Note: The best-performing value in each category is shown in bold, and the second-best is underlined. P-values are calculated relative to DPGOK.

Table S5. The median of AUPR for GO terms within each IC group

Methods	IC < 3			IC 3-5			IC > 5		
	MFO	BPO	CCO	MFO	BPO	CCO	MFO	BPO	CCO
BlastKNN	0.605	<u>0.091</u>	0.278	<u>0.770</u>	0.042	0.331	<u>0.504</u>	0.004	0.250
TALE	0.330	0.007	0.021	0.228	0.007	0.020	0.200	0.008	0.035
DeepGOZero	0.574	0.049	0.208	0.583	0.029	0.261	0.500	0.017	0.316
ATGO	0.500	0.010	0.221	0.337	0.002	0.089	0.012	0.002	0.012
DeepGO-SE	<u>0.625</u>	0.079	<u>0.383</u>	0.604	<u>0.065</u>	<u>0.347</u>	0.500	<u>0.091</u>	<u>0.440</u>
DPGOK	**0.693**	**0.141**	**0.450**	**0.798**	**0.111**	**0.445**	**0.611**	**0.111**	**0.500**
TALE+	0.513	<u>0.113</u>	0.250	0.507	<u>0.048</u>	0.358	<u>0.500</u>	<u>0.012</u>	0.266
ATGO+	<u>0.595</u>	0.111	<u>0.333</u>	<u>0.833</u>	<u>0.048</u>	<u>0.378</u>	<u>0.500</u>	0.004	<u>0.333</u>
DPGOK+	**0.745**	**0.168**	**0.457**	**0.923**	**0.132**	**0.476**	**0.833**	**0.103**	**0.500**

Note: The best-performing value in each category is shown in bold, and the second-best is underlined.

Table S6. Runtime and computational cost comparison of different methods on three sub-ontologies

cc	**GPU Mem (GiB reserved)**			**CPU RSS (GiB)**			**Total train time (min)**		
	MFO	**BPO**	**CCO**	**MFO**	**BPO**	**CCO**	**MFO**	**BPO**	**CCO**
BlastKNN	/	/	/	0.11	0.12	0.12	12	38	18
TALE	/	/	/	4.3	4.8	3.4	357	859	366
DeepGOZero	3.6	3.8	3.8	14.1	26.9	7.9	630	934	814
ATGO	/	/	/	10.3	35.1	9.0	10	29	12
DeepGO-SE	8.1	8.0	8.3	6.0	14.3	5.2	2332	3479	2846
DPGOK	7.0	8.7	4.4	3.8	9.3	3.0	23	128	21

Note: The reported training times exclude the feature extraction process from pre-trained models (such as ESM2 for DPGOK and DeepGO-SE) as well as the evaluation time. The training times for TALE and ATGO are based on CPU performance, as these methods lack GPU support.

Table S7. Performance of DPGOK on uncharacterized proteins

Protein name	**Homology GO num**	**Predicted GO num**	**Overlap GO num**	**Consistency**
A0A1U9X79	11	11	11	1.00
A4D1U5	9	9	9	1.00
D0PNI2	14	14	14	1.00
Q6IZA2	7	7	7	1.00
Q5T7S2	37	37	36	0.95
Q547U9	32	32	31	0.94
A0A1Y1BBN3	4	5	4	0.80
A0A2X0TVZ9	11	14	11	0.79
A0A7U3JW18	10	13	10	0.77
F6KCZ5	20	27	20	0.74

Note: 'Homology GO' denotes the functional annotations of uncharacterized proteins inferred from homologous proteins, and 'Consistency' is calculated using the Jaccard similarity

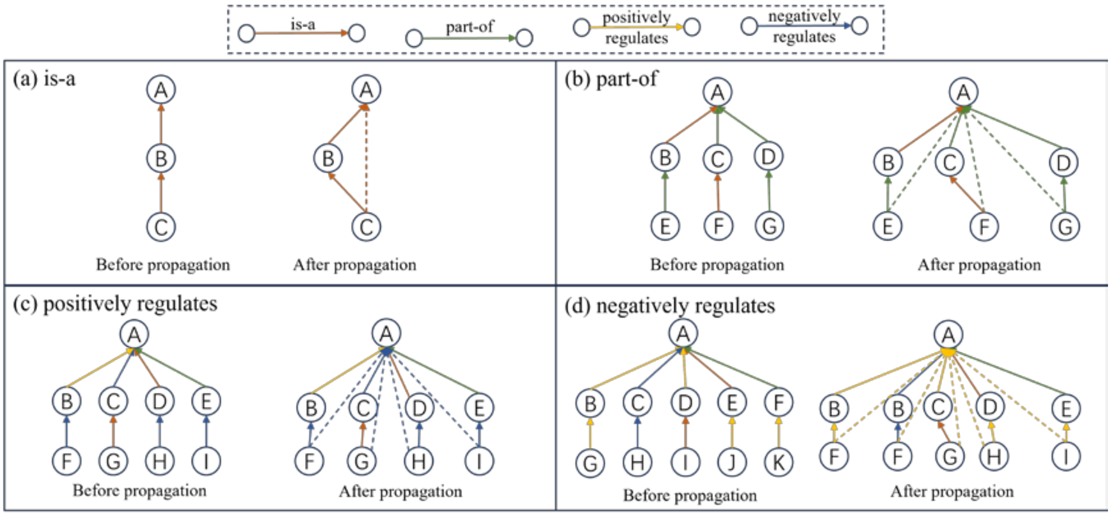

Fig. S1. Propagation strategies for each type of GO relation. The solid lines represent the original edges before propagation, while the dashed lines represent the newly established edges after propagation.

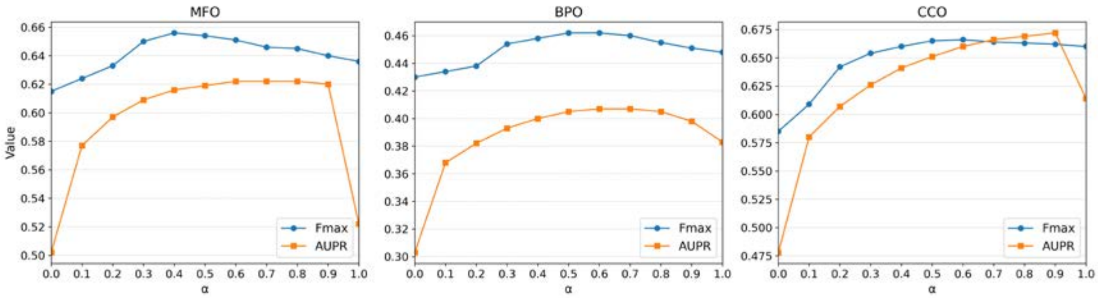

Fig. S2. The performance of Fmax and AUPR on Validation set with different fusion coefficients (α)

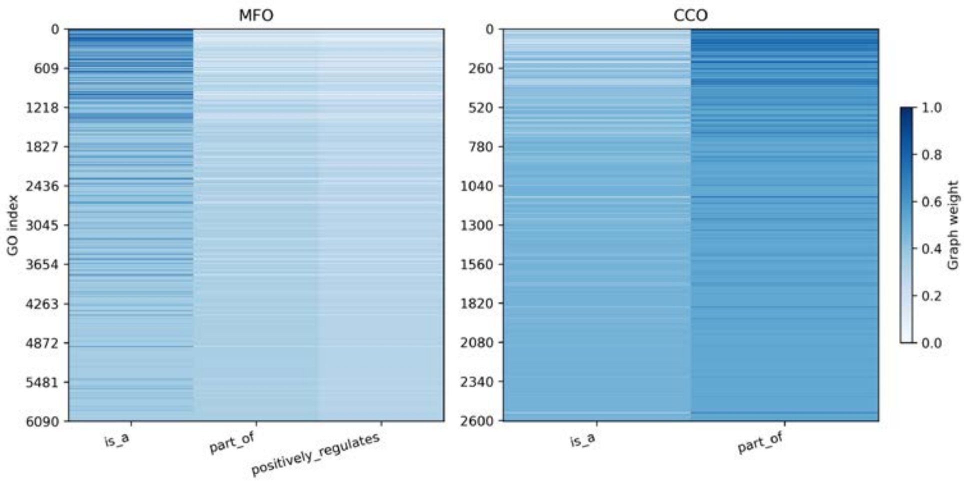

Fig. S3. Distribution of relation-specific weights for GO terms in MFO and CCO.